One World for the Platonic Solids

every mix of the five solids, with the rhombic dodecahedron and the cuboctahedron, on one cubic grid

DICTO (artist and geometer, Osaka, Japan) · dictospheres.com · contact@dictospheres.com · ORCID 0009-0001-2620-2036
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Only the cube fills space alone (the regular tetrahedron does not, against Aristotle: Lagarias and Zong 2012; Senechal 1981), and among pairs of regular solids only the tetrahedron with the octahedron does it (Conway and Torquato 2006; Conway, Jiao and Torquato 2011). #47 showed that, with a small kit of connecting pieces, every mix of the five Platonic solids fills space, built from four separate layers. Here the solids are allowed their own sizes, and something simpler appears: every solid fits inside a *skin* that turns it into a dodecahedron, and then every mix lives on one cubic grid, with no layers at all.

1. Free scale: two worlds, one seam

Theorem 1. With the dodecahedron of edge 1, the cube, tetrahedron and octahedron of edge φ and the icosahedron of edge √2/φ, each of the 31 mixes of the five Platonic solids fills space in two worlds: the cubic grid of edge φ (dodecahedra, Dogstars, cubes) and one octet of edge φ (tetrahedra, octahedra, and icosahedra in Pacioli's position with six Admiral Caps each). The two meet in one flat seam, face to face, and the connectors form a kit of six.

Turning the octet by 45° makes its square layers coincide with the cube faces of the grid. #47's fourth connector Q′ disappears, and with cubes in the mix the edging next to every Dogstar becomes a whole cube (cubes as cubes). Checked for all 31 mixes by random points: every point lies in exactly one piece.

2. One world: every solid in a skin

Theorem 2. Each of the five Platonic solids, and the cuboctahedron, fits inside a skin of congruent pieces that makes it exactly a dodecahedron of edge 1 (Table 1); the rhombic dodecahedron is a cube of edge φ with six square pyramids. Hence every mix of these seven solids (127 mixes) fills space in one world: the alternating cubes of the cubic grid of edge φ hold dodecahedral cells and rhombic dodecahedra, the cells between are Dogstar-type remainders, and the edging is X, XR and whole cubes. No seam remains.



Figure 1. The five solids in their skins (one-world sizes): the dodecahedron is its own cell; the icosahedron, cube, tetrahedron and octahedron sit inside glass dodecahedral cells of edge 1.

Table 1. The skins (the dodecahedral cell has volume $(15 + 7\sqrt{5})/4 \approx 7.663$; all values exact).

Solid inside	Edge	Volume of the solid	Volume of the skin	The skin is made of
Cube	φ	$2 + \sqrt{5} \approx 4.236$	$(7 + 3\sqrt{5})/4 \approx 3.427$	6 roofs
Tetrahedron	$\varphi\sqrt{2}$	$(2 + \sqrt{5})/3 \approx 1.412$	$(37 + 17\sqrt{5})/12 \approx 6.251$	4 corner tetrahedra (legs φ) + 6 roofs
Octahedron (small)	$\varphi/\sqrt{2}$	$(2 + \sqrt{5})/6 \approx 0.706$	$(41 + 19\sqrt{5})/12 \approx 6.957$	8 octant pieces + 6 roofs
Octahedron (sphere-unit)	$\varphi^2/\sqrt{2}$	$(9 + 4\sqrt{5})/6 \approx 2.991$	$(27 + 13\sqrt{5})/12 \approx 4.672$	8 congruent caps
Cuboctahedron	$\varphi/\sqrt{2}$	$5(2 + \sqrt{5})/6 \approx 3.530$	$(25 + 11\sqrt{5})/12 \approx 4.133$	8 corner tetrahedra (legs $\varphi/2$) + 6 roofs
Icosahedron (Karma)	$1/\varphi$	$5(\sqrt{5} - 1)/12 \approx 0.515$	$(25 + 8\sqrt{5})/6 \approx 7.148$	12 pentagonal pyramids + 30 AXE + 20 CLEO (→ #22)
Icosahedron (sphere-unit)	1	$5(3 + \sqrt{5})/12 \approx 2.182$	$(15 + 8\sqrt{5})/6 \approx 5.481$	6 Admiral Caps + the 8 caps
Dodecahedron	1	$(15 + 7\sqrt{5})/4 \approx 7.663$	0	none

The cube of edge φ is Euclid's cube inside the dodecahedron (Elements XIII.17), and the cube, its tetrahedron and the dodecahedron share one circumsphere (radius $\varphi\sqrt{3}/2 \approx 1.401$). The sphere-unit octahedron has its corners at the midpoints of the dodecahedron's six roof ridges; Pacioli's icosahedron of edge 1 sits inside it, its corners on the octahedron's edges at the golden section and on the faces of Euclid's cube.

Table 2. The kit. Beside the solids, the whole one world is built from these pieces (exact volumes).

Piece	Exact volume	≈	Used
Roof (Euclid's)	$(7 + 3\sqrt{5})/24$	0.5712	skins of the cube, tetrahedron, cuboctahedron
Cap (one shape)	$(27 + 13\sqrt{5})/96$	0.5841	8 round the sphere-unit octahedron, and round the icosahedron's octahedron
Admiral Cap	$(1 + \sqrt{5})/24 = \varphi/12$	0.1348	6 round the icosahedron in its octahedron (→ #21, #47)
Corner tetrahedron, legs φ	$(2 + \sqrt{5})/6$	0.7060	4 round the tetrahedron
Corner tetrahedron, legs $\varphi/2$	$(2 + \sqrt{5})/48$	0.0883	8 round the cuboctahedron
Dogstar	$(1 + \sqrt{5})/4 = \varphi/2$	0.8090	the cells between the dodecahedral cells (→ #42)
X (cube minus a roof)	$(41 + 21\sqrt{5})/24$	3.6649	edging beside a dodecahedral cell
XR (cube minus an RD pyramid)	$5(2 + \sqrt{5})/6$	3.5301	edging beside a rhombic dodecahedron
Y (cube plus a roof)	$(55 + 27\sqrt{5})/24$	4.8073	edging only in mixes without cubes (with cubes, whole cubes stand there)

Almost everything is a cube and a roof: a dodecahedron is a cube with six roofs, a Dogstar a cube without six, X a cube without one, Y a cube with one. Euclid's single construction carries the whole world.

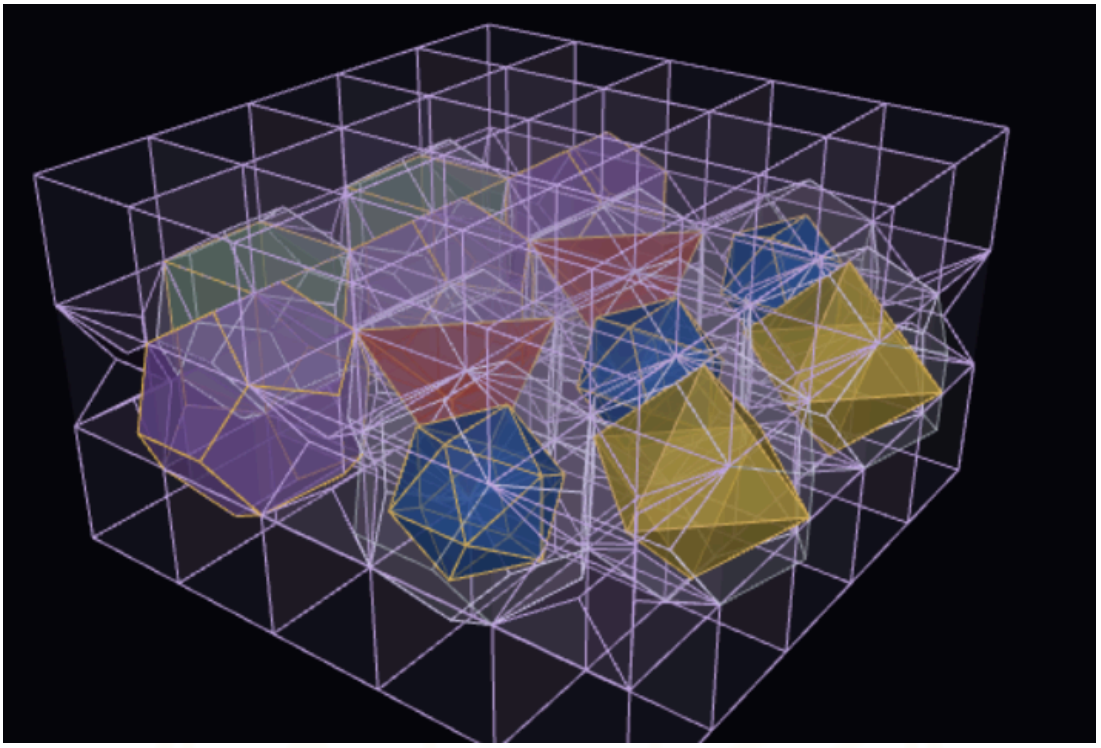


Figure 2. A block of the one world with all five solids (View in app: Every Mix of the Platonic Solids, One world).

3. The dodecahedron meets the icosahedron

Two exact bridges. **Karma cells:** by the Chain of Karma (→ #22), a dodecahedron of edge 1 is an icosahedron of edge 1/φ in a shell of 12 pentagonal pyramids, 30 AXE and 20 CLEO; such cells stand on the same grid as plain dodecahedra, so the two solids share one world.

> **Theorem 3 (the flat Karma layer).** The plane perpendicular to a three-fold axis at level φ/(2√3) above a layer of cell centres cuts every Dogstar it crosses so that the part below is exactly a CLEO, its apex a dodecahedron corner. The octet of edge φ/√2 above the plane, with Pacioli's icosahedra (edge 1/φ, the mirror form) in its octahedra, puts an icosahedron face exactly on every CLEO base.

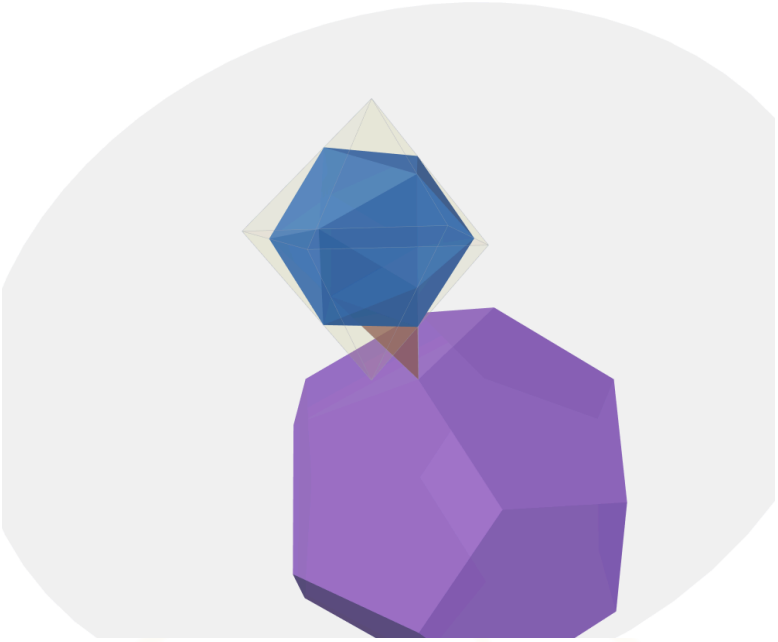


Figure 3. Icosahedron face → CLEO → dodecahedron corner across a flat plane (checked for all 60 CLEOs in a window: every one meets an icosahedron face exactly). The level φ/(2√3) is also the inradius of the Karma icosahedron.

4. How it was checked

Volumes and skins in exact arithmetic with √5; every filling by random points through the whole block, pieces as signed sums of convex parts, every point counted exactly once (all 31 and all 127 mixes); the face-to-face meetings by exact coordinates. Scripts and logs in the studies named above.

References

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Sources checked

Reference library: Euclid, Elements XIII (XIII.17, the cube in the dodecahedron); Pacioli, De divina proportione (the golden rectangles); Kepler, Mysterium cosmographicum (1596, nested solids); G. W. Hart on the Dogstar (stellation 8 of the dodecahedron, alternating with dodecahedra); DICTO #20, #21, #22, #42, #44, #47. Web: Torquato and co-workers on tilings by pairs of regular polyhedra (only cubes of different sizes, or tetrahedra with octahedra); Dense packings of Platonic and Archimedean solids (arXiv 0909.0940); no space-filling of all five Platonic solids together was found.

DICTO: ideas, names, constructions and direction. Computation, verification and drafting with Claude (Anthropic).